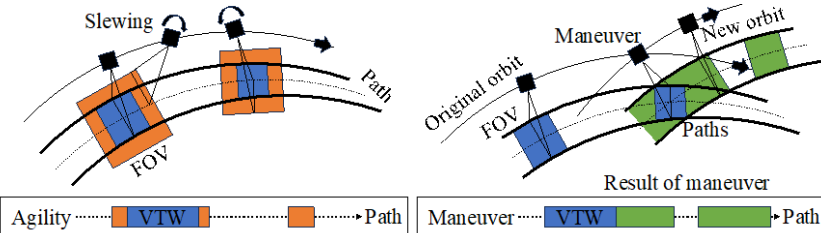
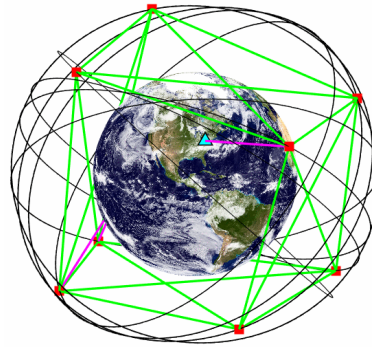


Mission Design and Operations

- Autonomous operations under uncertainty
- Mission design and scheduling
- Network fault detection & resilience



New Earth observing strategies

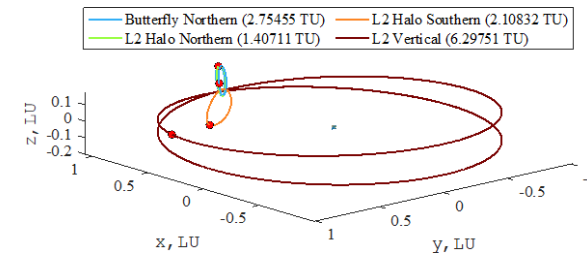


Satellite networks

Space Traffic Management & Space Domain Awareness

- Post mission disposal
- Space-based lasers
- PNT

Active debris removal

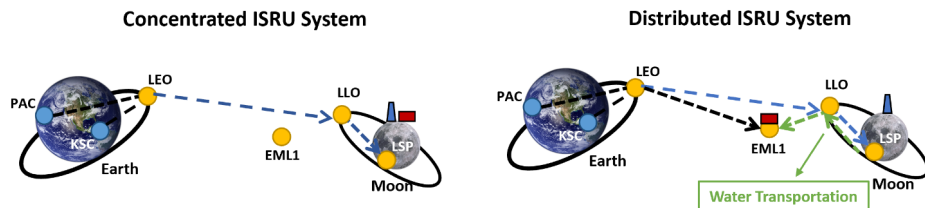


Cislunar SDA architecture design

Distributed Satellite Systems

On-orbit Servicing & Space Logistics

- *Where do we locate gas stations in space?*
- *In-situ* resource utilization
- Campaign-level mission design & supply chain mgmt.



KSC: Kennedy Space Center
PAC: Pacific Ocean
LEO: Low-Earth Orbit
LSP: Lunar South Pole
LLO: Low-Lunar Orbit
EML: Earth-Moon Lagrangian Points

Soil Water Extraction (SWE) ISRU
Direct Water Electrolysis (DWE) ISRU
Earth Surface Node
Earth/Cis-Lunar System Node

Cislunar space logistics

Operations Research

➤ The science of making “good” decisions

Tools

- Modeling & Simulation
 - Graph-theoretic
 - Network analysis
- Mathematical optimization
 - Convex, MINLP, large-scale, decomposition-based
- AI, ML, Computer Vision

